

Self-Renewing Antifouling Coating Based on Integrated Offensive and Defense Strategies Inspired by Coral

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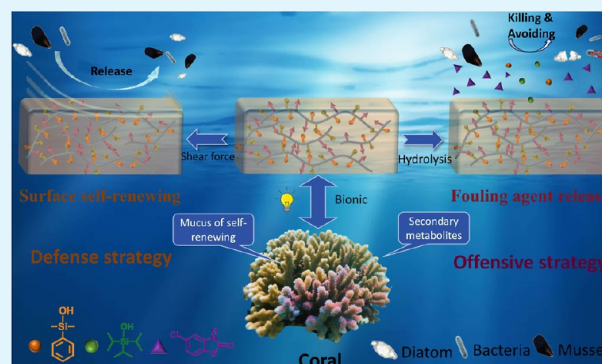
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Supporting Information

ABSTRACT: Silane acrylate self-polishing antifouling coatings play a crucial role in environmentally friendly biological fouling prevention. However, they still face challenges related to static fouling conditions and limited service life. In this study, inspired by the dual antifouling mechanism of coral, mucus is “self-renewing” and actively releases antifouling substances. A surface self-renewing antifouling coating (MEIPT-*x*) was prepared based on two hydrolyzable silane side chains that accelerate surface self-renewal, along with an eco-friendly antifouling agent, 5-chloro-2-benzothiazolinone (TL), grafted as a dangling group. This coating demonstrates a controllable hydrolysis rate that accelerates with an increasing content of hydrolyzable monomers. The synergistic effect of the “defense” and “offensive” of the MEIPT-*x* coating enables effective antibacterial (94.8% for marine bacteria), antidiatom (98.6% for *Halamphora* sp. and 99.8% for *Nitzschia closterium*), and antimussel properties. In addition, the excellent antifouling performance of the MEIPT-*x* coating can be sustained for over 11 months under real marine conditions. Consequently, this integrated offensive and defensive strategy against biofouling may offer promising perspectives for applications in ship antifouling.

KEYWORDS: antifouling coating, self-renewing, benzothiazolinone, antibacterial, bioinspired



1. INTRODUCTION

The attachment of marine fouling organisms can significantly increase the navigation resistance of vessels, exacerbate fuel consumption, promote microbial corrosion, and potentially lead to biological invasions.^{1–4} Antifouling coatings are among the most economical, efficient, and practical methods for preventing marine biological fouling.^{5–7} Due to the severe damage caused by tributyltin to marine ecosystems, its use in ship coatings has been prohibited.^{8–10} This development underscores the urgent need for eco-friendly methods to prevent biofouling. In recent years, coatings based on biomimetic strategies have attracted widespread attention for their environmentally friendly antifouling performance.^{11,12}

“Natural selection, survival of the fittest”. The diverse biological structures and phenomena that have evolved through natural selection have inspired the development and improvement of advanced materials.^{13,14} Munther et al. simulated shark skin surfaces using various microfabrication and microforming techniques to produce planar scale patterns with engineered height gradients designed to prevent biofouling. The resulting surface demonstrated significant antibacterial activity and excellent wear resistance.¹⁵ Tong et al. drew inspiration from the defensive behavior of sea anemones and constructed the “4D SLIPS” (FSLIPS)

antifouling material. FSLIPS regulates the movement of the cilia through an external magnetic field and controls the release of bactericidal nitric oxide through light response, resulting in excellent antifouling properties.¹⁶ However, although these biomimetic materials exhibit excellent antifouling properties, their preparation processes are complex and challenging to scale up. Therefore, there is an urgent need to develop a simple and efficient antifouling coating to address the problem of biofouling.

Coral is a sessile organism that cannot move even in response to external stimuli, making it necessary to apply specialized antifouling strategies.¹⁷ Antifouling coatings inspired by these strategies can provide effective protection for ships and structures in static underwater environments.^{18,19} The mucus layer secreted by the coral surface can detach and remove fouling organisms.²⁰ Additionally, corals can produce a variety of secondary metabolites, which possess antibacterial

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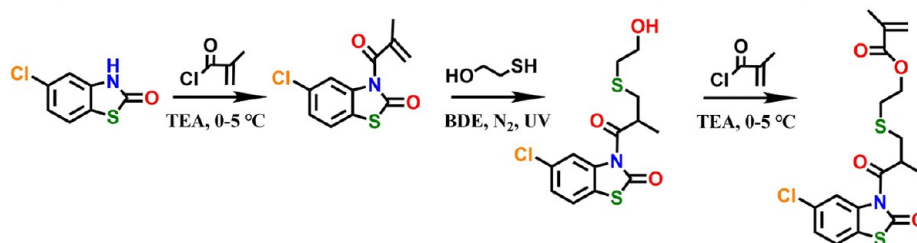
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a Synthesis of benzothiazolinone-based methyl methacrylate monomer



b Synthesis of benzothiazolinone-based acrylic silane resin

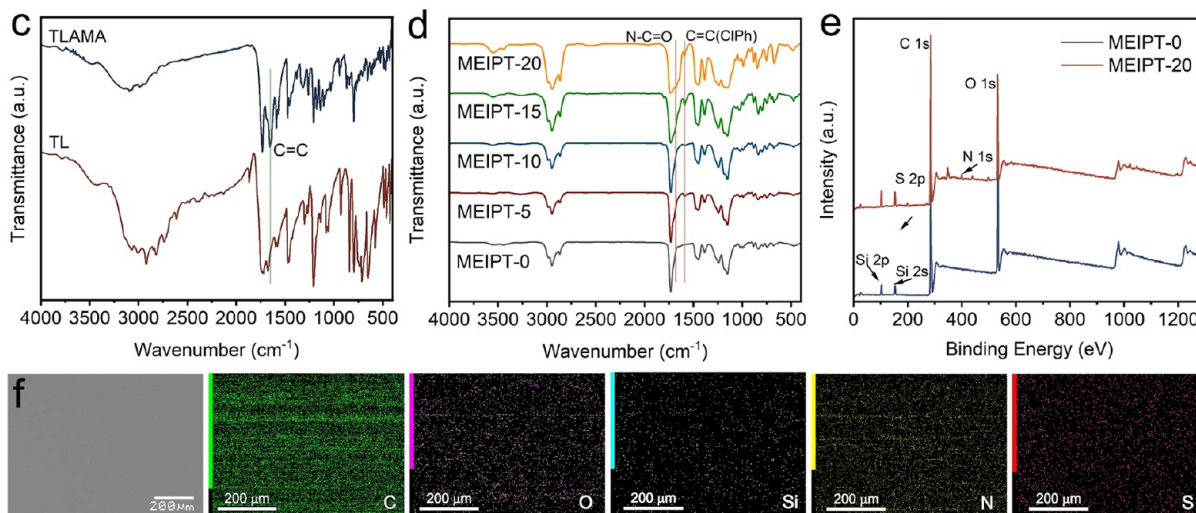
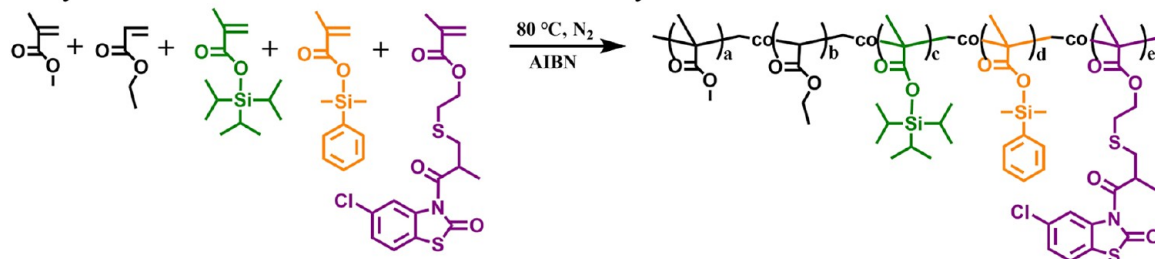


Figure 1. The synthesis diagram of (a) TLAMA monomer and (b) MEIPT-*x* copolymers; FT-IR spectra of (c) TLAMA and (d) MEIPT-*x*; (e) XPS spectra of MEIPT-0 and MEIPT-20; and (f) SEM image and corresponding mapping images of MEIPT-20.

properties and resistance to algal growth and larval adhesion.^{17,21} Inspired by the coral “defense and offensive” dual antifouling strategy, an eco-friendly self-renewing antifouling coating (MEIPT-*x*) was developed for marine antifouling. By adjustment of the content of the two hydrolyzed side chains of triisopropylsilyl methacrylate (TISMA) and phenyldimethylsilyl methacrylate (MPSMA), the self-renewing rate of the coating surface in static environments can be regulated, thereby improving its physical defense capabilities. Simultaneously, 5-chloro-2-benzothiazolinone (TL), chemically bonded to the coating, can achieve controlled release, creating an antifouling environment around the coating and actively killing attached bacteria and diatoms. Benzothiazole derivatives are recognized as promising marine antifouling agents due to their ability to disrupt the cellular structure of microorganisms, exhibiting strong antibacterial and algicidal properties.²² MEIPT-*x* coatings achieve efficient and long-lasting antifouling performance by regulating the self-renewing rate of its surface and releasing the antifouling active substance TL through a dual strategy of “defense and offensive”. It offers theoretical and technical assistance for

the advancement of promising self-renewing antifouling coating.

2. EXPERIMENTAL SECTION

2.1. Materials.

Details are listed in [Supporting Information](#).

2.2. Preparation of TLAMA. TL (1 g) and triethylamine (TEA, 0.82 g) were predissolved in tetrahydrofuran (THF) and stirred in an ice–water bath. Methacryloyl chloride (0.57 g) was dissolved in THF and added to a pressure funnel. After the dropwise addition was completed, the reaction was carried out overnight at room temperature. The TEA hydrochloride produced by the reaction was removed by filtration, and THF and excess TEA were removed by rotary evaporation to obtain benzothiazolinone-based methyl acrylamide (TLA). Subsequently, the obtained TLA (1 g), β-mercaptoethanol (0.84 g), and benzoin dimethyl ether (2 drops) were dissolved in THF. The mixture was stirred under N₂ protection and UV light irradiation for 3 h, and the product was precipitated in distilled water 3 times. After vacuum drying at 60 °C for 24 h, a hydroxyl-modified benzothiazolinone-based amide (TLAM) was obtained. Finally, the obtained TLAM (1 g) and TEA (0.45 g) were predissolved in THF and stirred in an ice–water bath. Methacryloyl chloride (0.31 g) was dissolved in THF and added to a constant pressure funnel. After the dropwise addition was complete,

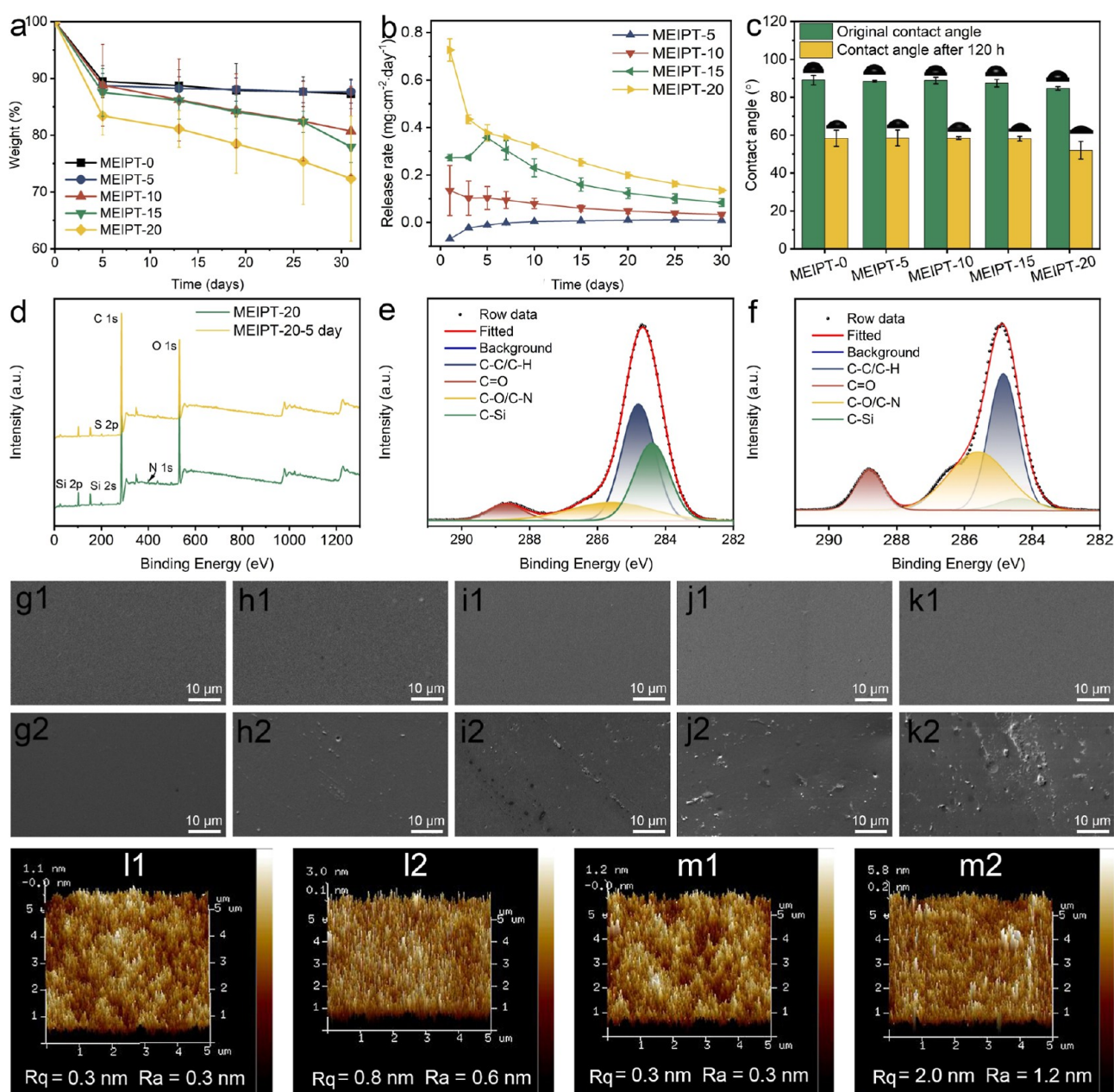


Figure 2. (a) Weight loss rate of MEIPT-*x* coatings; (b) release rate of TL from MEIPT-*x* coatings in ASW water at room temperature; (c) the contact angle of MEIPT-*x* coatings before and after immersion in ASW for 120 h; (d) XPS full spectra of MEIPT-20 and MEIPT-20-5 days; high-resolution C 1s spectrum of (e) MEIPT-20 and (f) MEIPT-20-5 days; SEM images of MEIPT-*x* coatings (g1–k1) before and (g2–k2) after 5 days of immersion in ASW; AFM images of MEIPT-0 and MEIPT-20 (l1–m1) before and (l2–m2) after 5 days of immersion in ASW. Note: the error bars represent standard deviations.

the reaction was carried out overnight at room temperature. The TEA hydrochloride produced by the reaction was removed by filtration, and THF and excess TEA were removed by rotary evaporation to obtain the benzothiazolinone-based methyl methacrylate monomer (TLAMA).

2.3. Synthesis of MEIPT-*x*. MEIPT-*x* was prepared by using free radical polymerization. First, a mixed solution of DMF and propylene glycol methyl ether (PGME, 30 g, DMF/PGME = 2:1, mass ratio) was added to a four necked flask equipped with a stirrer, condenser, dropper funnel, and thermometer and heated to 80 °C. Subsequently, methyl methacrylate (MMA), ethyl acrylate (EA), TISMA, MPSMA, TLAMA, azobis(isobutyronitrile) (AIBN), and the mixed solution of DMF and PGME (70 g) were thoroughly mixed in proportion and slowly dripped into the flask under N₂ protection for more than 3 h. Subsequently, the mixed solution of DMF and PGME (20 g) and

benzoyl peroxide (0.32 g) were added dropwise to the flask and stirred for another 2 h. Finally, the polymer was exposed to air to terminate the reaction to obtain MEIPT-*x* (Figure 1b). The preparation process for the self-renewing antifouling coating system is shown in Figure S1. The dosing ratios of the monomers and molecular weights (Mw) used in various resins are shown in Table S1.

2.4. Characterization. Details are listed in Supporting Information.

3. RESULTS AND DISCUSSION

3.1. Characterization of TLAMA and MEIPT-*x*. The MEIPT-*x* coating is designed to enhance the static antifouling performance and extend the antifouling duration of silicon-based coatings by utilizing two hydrolyzable silane monomers

and grafting eco-friendly antifouling agents (Figure 1b). MMA and EA can modify the mechanical properties, film-forming ability, and weather resistance of the coatings. TISMA and MPSMA side chains can regulate the hydrolysis rate of coatings, enabling long-term and stable surface renewal. By controlling the hydrolytic release of the dangling groups grafted with TL, fouling organisms can be effectively killed and prevented. This is the coordinated integration of physical defense and chemical offensive antifouling strategies. In addition, the environmental friendliness of MEIPT-*x* coatings is discussed in the Supporting Information under section "S2.4 Environmentally Friendly Performance of MEIPT-*x* Coatings".

The molecular structure of the antifouling TLAMA monomer was verified by ¹H NMR spectroscopy (Figure S2). Characteristic peaks were observed at 7.75–7.56, 5.86, 4.16, 4.00, 2.89, and 1.07 ppm, corresponding to the benzene ring, double bond, and methylene and methyl groups at different positions in the TLAMA structure. The FT-IR spectra of TL and TLAMA are shown in Figure 1c. The new peak of TLAMA at 1650 cm⁻¹ is the stretching vibration of C=C bonds in ethylene groups.²³ These results indicate the successful preparation of TLAMA.

The FT-IR spectra of MEIPT-*x* coatings are shown in Figure 1d. Compared to MEIPT-0, other MEIPT-*x* coatings exhibit significant new peaks at 1595 cm⁻¹ and 1683 cm⁻¹, representing the C=C skeleton vibration in halogenated benzene rings and the stretching vibration of N–C=O bonds, respectively. This is attributed to the presence of chlorinated benzene rings and amide bonds in TLAMA monomers, which are absent in MEIPT-0, confirming the incorporation of the antifouling monomer TLAMA. The XPS spectra of MEIPT-0 and MEIPT-20 are shown in Figure 1e. Compared with MEIPT-0, MEIPT-20 exhibits new S and N characteristic peaks. In addition, its SEM-mapping shows that C, N, O, S, and Si elements are uniformly distributed on the surface of the coating (Figure 1f). These results indicate that MEIPT-*x* coatings have been successfully prepared.

3.2. Hydrolysis Properties. A stable static polishing rate is one of the challenges faced by self-polishing coatings (SPCs) in practical applications.²⁴ To assess the degradation capability of the MEIPT-*x* coatings in static seawater, we conducted a one month artificial seawater (ASW) immersion weight loss experiment (Figure 2a). The weight loss rate of the MEIPT-*x* coatings first increases rapidly and then gradually reaches a dynamic equilibrium. In addition, the weight loss rate of the coating is also positively correlated with the content of hydrolyzable monomers. The weight loss rate of the coating is calculated using eq S1. Compared with 12.7% of MEIPT-0, the weight loss rate of MEIPT-20 increased to 27.7% after 30 days. The result shows that the coating with a controllable self-renewing rate can be designed by adjusting the content of the hydrolyzable side chain. The release behavior of antifouling active substances directly determines the antifouling effectiveness of the self-renewal antifouling coating. Figure 2b shows the release rate of TL. Compared to 0.008 mg cm⁻² day⁻¹ of MEIPT-5, the release rate of MEIPT-20 reaches 0.135 mg cm⁻² day⁻¹ after 30 days of immersion. The content of the MEIPT-20 hydrolysis monomer is relatively appropriate. However, if the hydrolysis monomer content is too high, it can cause a significant increase in the initial release of the antifouling agent, posing environmental risks. Additionally, the degradation rate may far exceed the surface renewal rate, causing the coating to expand excessively and form a loose,

porous structure, which increases the attachment area for marine organisms. Therefore, as the immersion time increases, the MEIPT-20 coating rapidly stabilizes, and its self-renewal process can be considered nearly uniform and gradual, which helps ensure consistent and long-lasting antifouling effects.²⁵

3.3. Surface Wettability and Roughness. Surface wettability and surface roughness play crucial roles in antifouling performance. Further study of the interaction between surface wettability and the hydrolysis kinetics of MEIPT-*x* coatings is conducted through static water contact angle (WCA) testing (Figure 2c). The average WCA of MEIPT-0 is 89.1°. As the TLAMA content increases and the TISMA content decreases, the WCA of the coating gradually decreases until drops to 84.7° of MEIPT-20. After 120 h of ASW immersion, the contact angle of the MEIPT-20 coating surface decreased to 52.0°. This is attributed to the hydrolysis of more side chains, which generates a greater number of hydrophilic carboxyl groups on the surface. For further verification, the composition and surface element content of MEIPT-20 before and after immersion in ASW for 5 days were analyzed by XPS (Figure 2d). XPS offers excellent capability for analyzing the chemical bonding states within the surface layer top 10 nm.²⁶ As shown in Figure 2e, the high-resolution spectra of C 1s of MEIPT-20 correspond to C–C/C–H, C–Si, C–O/C–N, and C=O peaks at 284.8, 284.3, 285.9, and 288.8 eV binding energies, respectively.^{27,28} For MEIPT-20–5 day, the peak areas of C–O and C=O bonds on the coating surface are larger than before hydrolysis, while the peak area of the Si–C bond is lower than before hydrolysis (Figure 2f). Meanwhile, the C/Si ratio increased from 10.6 to 19.9 after 5 days (Table 1). These results are attributed to a significant

Table 1. The Element Content of MEIPT-20 and MEIPT-20 (5 Days) Coating Surfaces

samples	XPS atom concentration (%)					
	C	O	Si	S	Cl	N
MEIPT-20	71.66	18.72	6.75	0.50	0.75	1.62
MEIPT-20–5 days	73.11	22.41	3.67	0.16	0.08	0.57

number of hydrophilic carboxyl groups being generated on the coating surface following the hydrolysis of TLAMA, TISMA, and MPSMA side chains. The hydrolysis of this side chain promotes a dual defense function: the formation of a fouling-release layer and an antifouling adhesion barrier on the coating surface.

The surface morphology of the coating before and after immersion in ASW is crucial. SEM images show that the surfaces of the MEIPT-*x* coatings are uniform and smooth before immersion (Figure 2g1–k1), indicating that the coating system possesses excellent film-forming properties. However, the surface of the MEIPT-*x* coating presents a relatively rough morphology after 5 days of ASW immersion, especially for MEIPT-20 (Figure 2g2–k2). In addition, the surface roughness of the MEIPT-*x* coating was further evaluated by AFM (Figure 2l,m). The Rq and Ra values of MEIPT-0 and MEIPT-20 were both 0.3 nm before immersion. The Rq and Ra of MEIPT-0 increased to 0.8 and 0.6 nm after 5 days of immersion in ASW, respectively. However, the Rq and Ra of MEIPT-20 increased to 2.0 and 1.2 nm under the same conditions, respectively. These results further prove that the hydrolysis rate of the coating increases with the increase of hydrolysis monomer content.

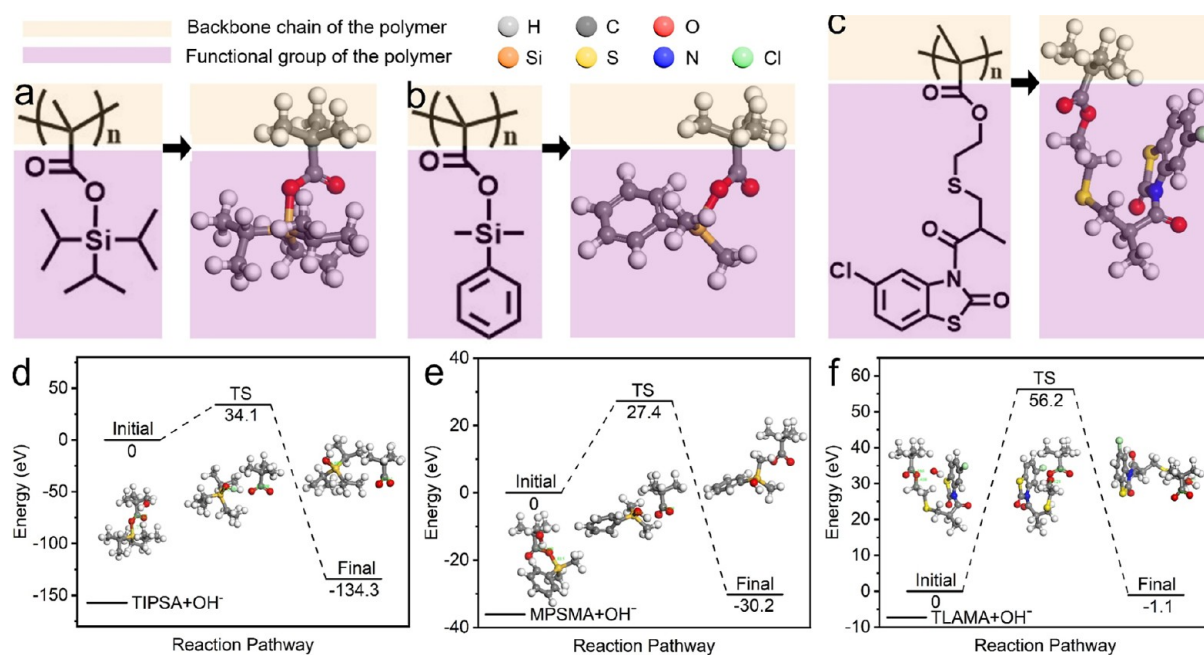


Figure 3. Molecular modeling and simplified backbone chain of: (a) TISMA, (b) MPSMA, and (c) TLAMA polymers for DFT calculations; relative energy as change of reaction pathways (adsorption state-transition state-final state) of (d) TISMA, (e) MPSMA, and (f) TLAMA molecules based on DFT calculations.

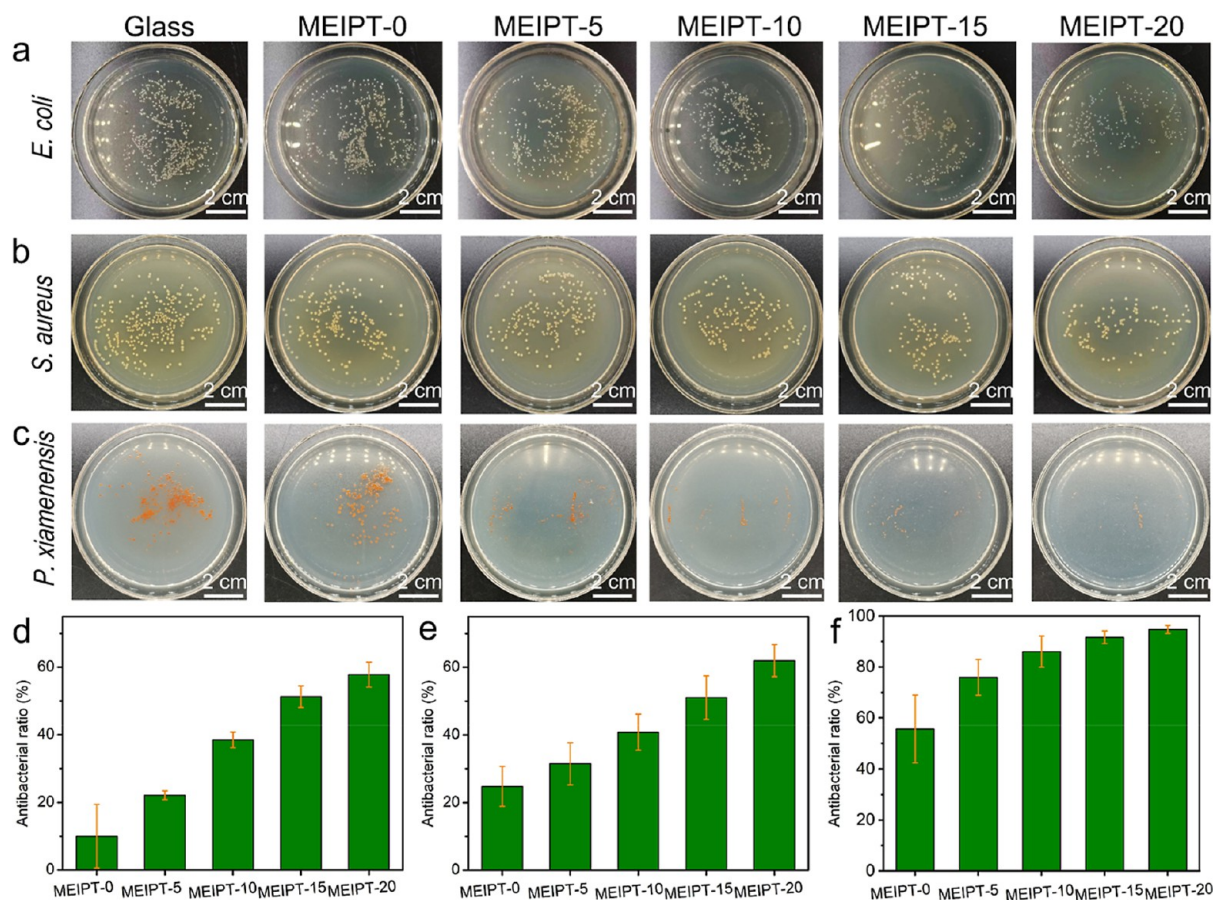


Figure 4. Digital photos of the antibacterial performance of MEIPT-*x* coating on (a) *E. coli*, (b) *S. aureus*, and (c) *P. xiamenensis*; the antibacterial efficiency of MEIPT-*x* coating against (d) *E. coli*, (e) *S. aureus*, and (f) *P. xiamenensis*. Note: the error bars represent standard deviations.

To gain a deeper understanding of the surface self-renewal mechanism of the coatings, density functional theory (DFT)

was employed to calculate and elucidate the hydrolysis mechanism of monomers. Due to the limitations of DFT in

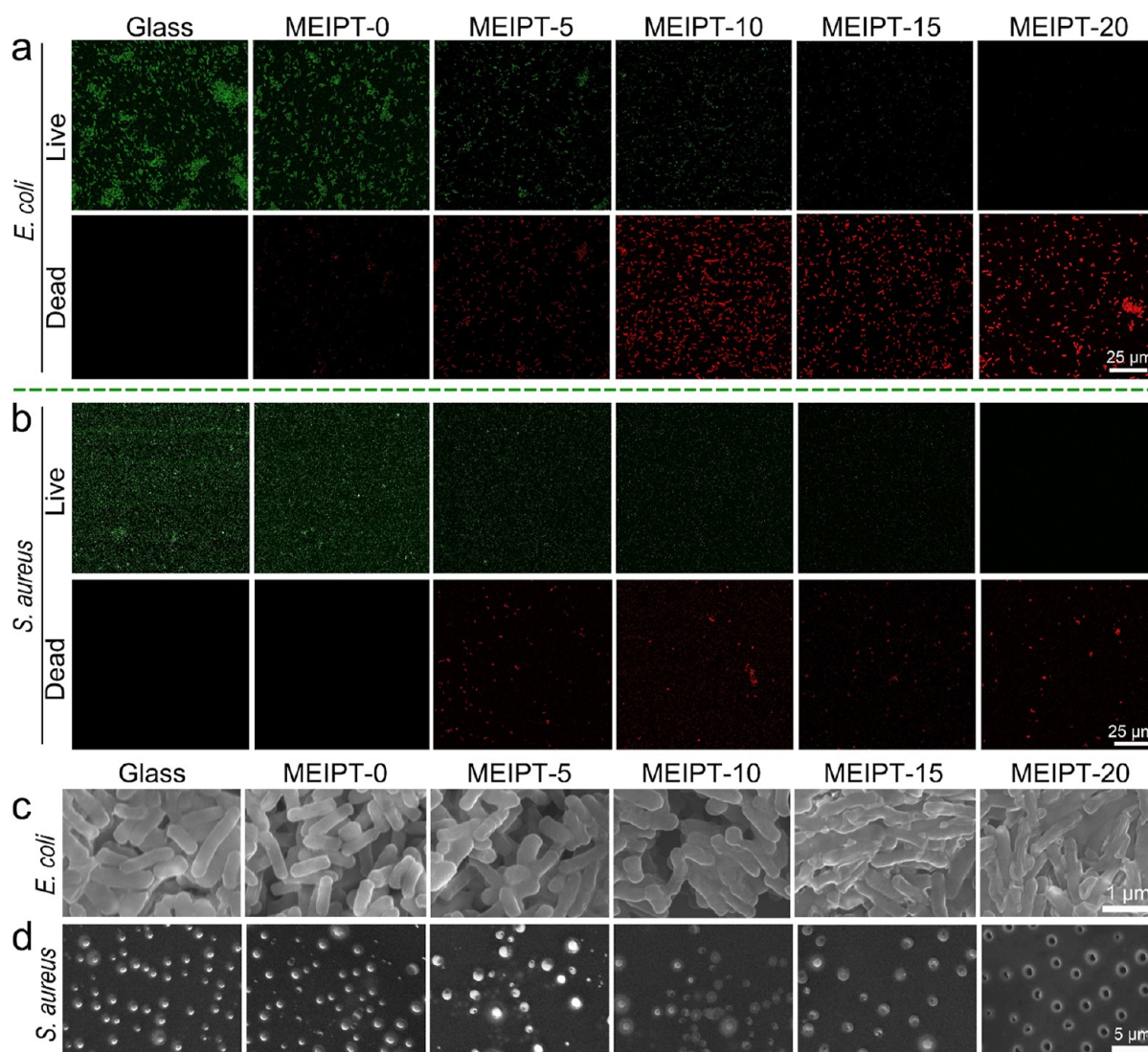


Figure 5. The confocal microscopy images of (a) *E. coli* and (b) *S. aureus* are attached to different coating surfaces. Use green fluorescence to represent live bacteria and red fluorescence to represent dead bacteria. The SEM images of (c) *E. coli* and (d) *S. aureus* are attached to different coating surfaces.

managing a large number of atoms, we simplified the TISMA, MPSMA, and TLAMA molecule designs. The hydrolysis side chain was streamlined to include only the dangling groups near the primary decomposition point. As illustrated in Figure 3a–c, the coating comprises a main chain (yellow portion) and hydrolyzed side chains (purple portion). The functional groups of the polymer side chains dictate their hydrolysis mechanisms and properties. To gain a deeper understanding of the relative energy changes during the decomposition of hydrolysis monomers, Figure 3d–f illustrates the variations in the relative energy of three hydrolysis monomers in relation to the reaction pathway. From the perspective of hydrolysis mechanisms, TIPSA and MPSMA exhibit similar behavioral patterns. The activation energy of the MPSMA molecule (27.4 eV) is lower than that of the TIPSA molecule (34.1 eV). This difference can be attributed to the fact that MPSMA exhibits lower overall steric hindrance compared to TIPSA, which possesses ductility and greater steric hindrance. This reduced steric hindrance in MPSMA facilitates the hydrolysis reaction. In addition, the activation energy of TLAMA molecules is relatively high, which can be attributed to a higher bond energy of the C–O

chemical bonds within TLAMA. Consequently, the hydrolysis reaction necessitates overcoming a greater energy barrier, leading to a slower hydrolysis rate.

3.4. Antifouling Properties. The ability to resist primary biofouling (bacteria) is essential, as the adhesion of bacteria contributes to the formation of biofilms.^{29,30} This biofilm creates favorable conditions for the subsequent attachment of additional marine fouling organisms.³¹ The antibacterial performance of MEIPT-*x* coatings is evaluated through adhesion tests involving three types of bacteria: one marine bacteria, *Pseudoalteromonas xiamenensis*, and two common bacterium, *Escherichia coli* and *Staphylococcus aureus* (Figure 4). Utilizing glass slides as the control group, we assessed the antibacterial efficacies of the coatings through cell counting. The optical photos of *E. coli*, *S. aureus*, and *P. xiamenensis* are presented in Figure 4a–c. It is observed that as the TL content increased, the attachment of the three types of bacteria to the coating surface progressively diminished, highlighting the antibacterial effect of TL. The antibacterial rate is calculated using eq S2. From Figure 4d–f, it can be observed that the antibacterial efficiency of MEIPT-20 against *E. coli*, *S. aureus*,

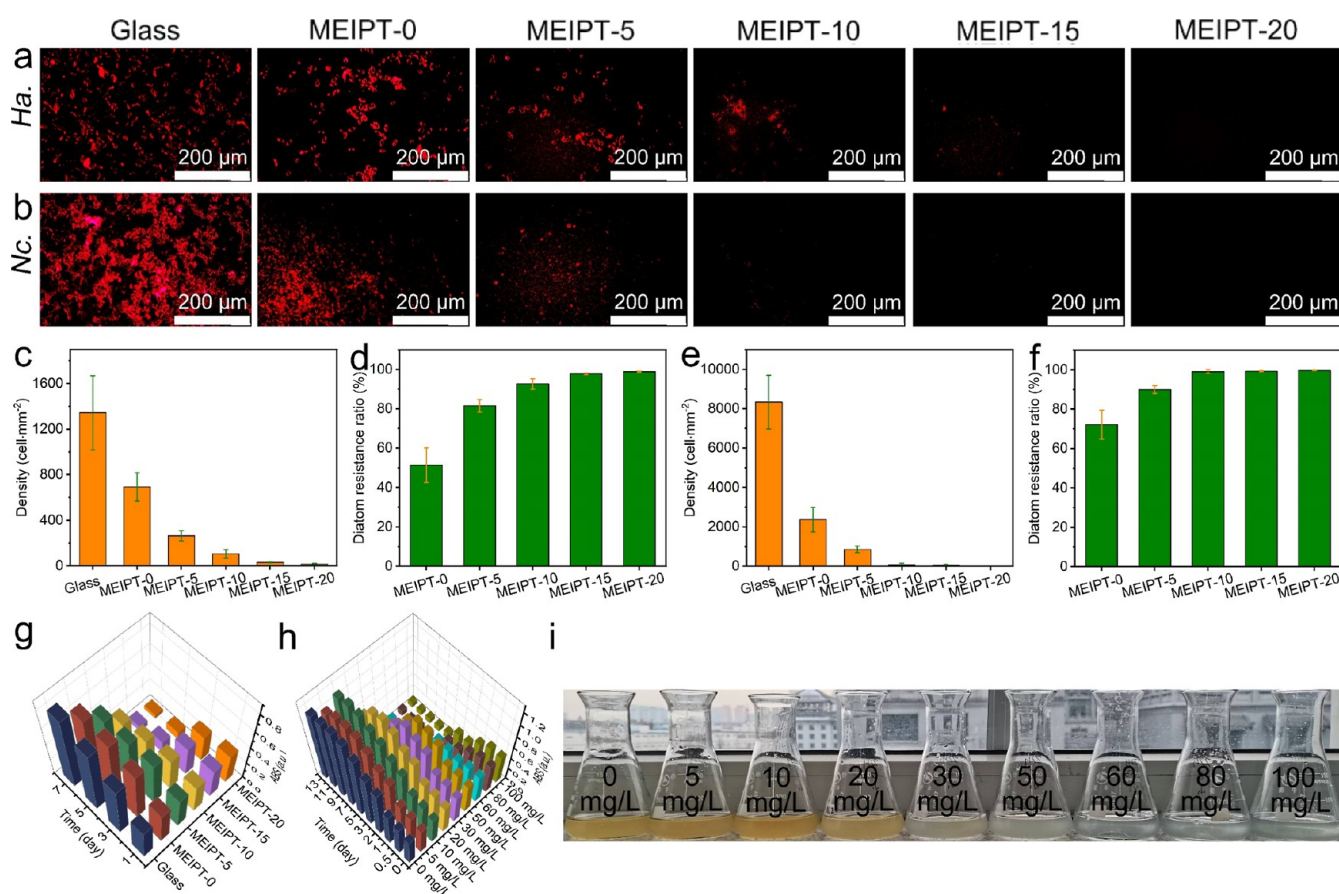


Figure 6. Fluorescence image of MEIPT-*x* coatings immersed in (a) *Ha.* and (b) *N. closterium* diatom solution after 7 days; (c) density and (d) adhesion inhibition ratio of *Ha.* diatoms on the surface of MEIPT-*x* coating; (e) density and (f) adhesion inhibition ratio of *N. closterium* diatoms on the surface of MEIPT-*x* coating; (g) toxicity evaluation of MEIPT-*x* coating on *N. closterium* diatom; (h) toxicity evaluation of different concentrations of TL on *N. closterium* diatom; and (i) digital photos after 2 weeks of testing. Note: the error bars represent standard deviations.

and *P. xiamenensis* is 57.8%, 62.0%, and 94.8%, respectively. This can be attributed to TL destroying bacterial cell membranes and leading to the leakage of important substances in cells, thus inhibiting bacterial proliferation.^{32–34} Compared to *E. coli* and *S. aureus*, the antibacterial rate of over 90% of *P. xiamenensis* marine bacteria can be attributed to their long-term adaptation to high-salinity and oligotrophic environments. Their cell membranes exhibit an increased fluidity, making them more susceptible to TL damage.

To further evaluate the bactericidal effect of MEIPT-*x* coatings, the survival ability of *E. coli* and *S. aureus* was evaluated using the standard live/dead method (Figure 5a,b). *E. coli* and *S. aureus* are observed to strongly adhere to the surfaces of glass and MEIPT-0 while maintaining their viability and exhibiting intense green fluorescence. On the contrary, the surface of the MEIPT-20 coating shows obvious red fluorescence, indicating its superior bactericidal properties. The coating surface containing TL dangling groups can effectively kill attached bacteria, and its antibacterial efficacy is dependent on the content of TL. As the TL content increases, the bactericidal effect is enhanced. Similarly, SEM images clearly revealed that TL can effectively disrupt the integrity of bacterial cell membranes, resulting in the leakage of cellular contents and ultimately leading to their death (Figure 5c,d).

Halophora sp. (*Ha.*, benthic diatoms) and *Nitzschia closterium* (planktonic diatoms) are used as typical fouling

organisms to further assess the antiadhesion performance of the MEIPT-*x* coatings against diatoms (Figure 6a,b). After 7 days of cultivation, both types of diatoms are found to strongly adhere to the surface of the glass. In stark contrast, the number of diatoms attached to the MEIPT-20 surface is very low, indicating its effective resistance to diatom adhesion. In addition, the adhesion density of diatoms and the inhibitory diatom adhesion rate of MEIPT-*x* coatings are shown in Figure 6c–f. The *Ha.* diatom density of the glass (1342 cell mm^{-2}) and MEIPT-0 (693 cell mm^{-2}) is significantly higher than that observed in MEIPT-20 (15 cell mm^{-2}). The anti-diatom adhesion rate is calculated using eq S3. Furthermore, the inhibition ratios of *Ha.* following a 7 day immersion period with MEIPT-0, MEIPT-5, MEIPT-10, MEIPT-15, and MEIPT-20 are recorded as 51.3%, 81.5%, 92.7%, 97.8%, and 98.6%, respectively. The increase in TL content within the coating has a significant repellent effect on benthic diatoms. In contrast, planktonic diatoms lack autonomous action ability and depend exclusively on the dynamics of water flow for their movement. Consequently, planktonic diatoms exhibited a greater adherence to the surfaces of both the glass (8337 cell mm^{-2}) and MEIPT-0 (2363 cell mm^{-2}). Notably, the surface diatom density of MEIPT-20 is recorded at only 19 cell mm^{-2} , while its inhibition ratio of *N. closterium* diatom reaches an impressive 99.8%. In addition, the growth curves of cocultured *N. closterium* with the MEIPT-*x* coatings are illustrated in Figure 6g. As the TL content in the coating surpasses 10%, the

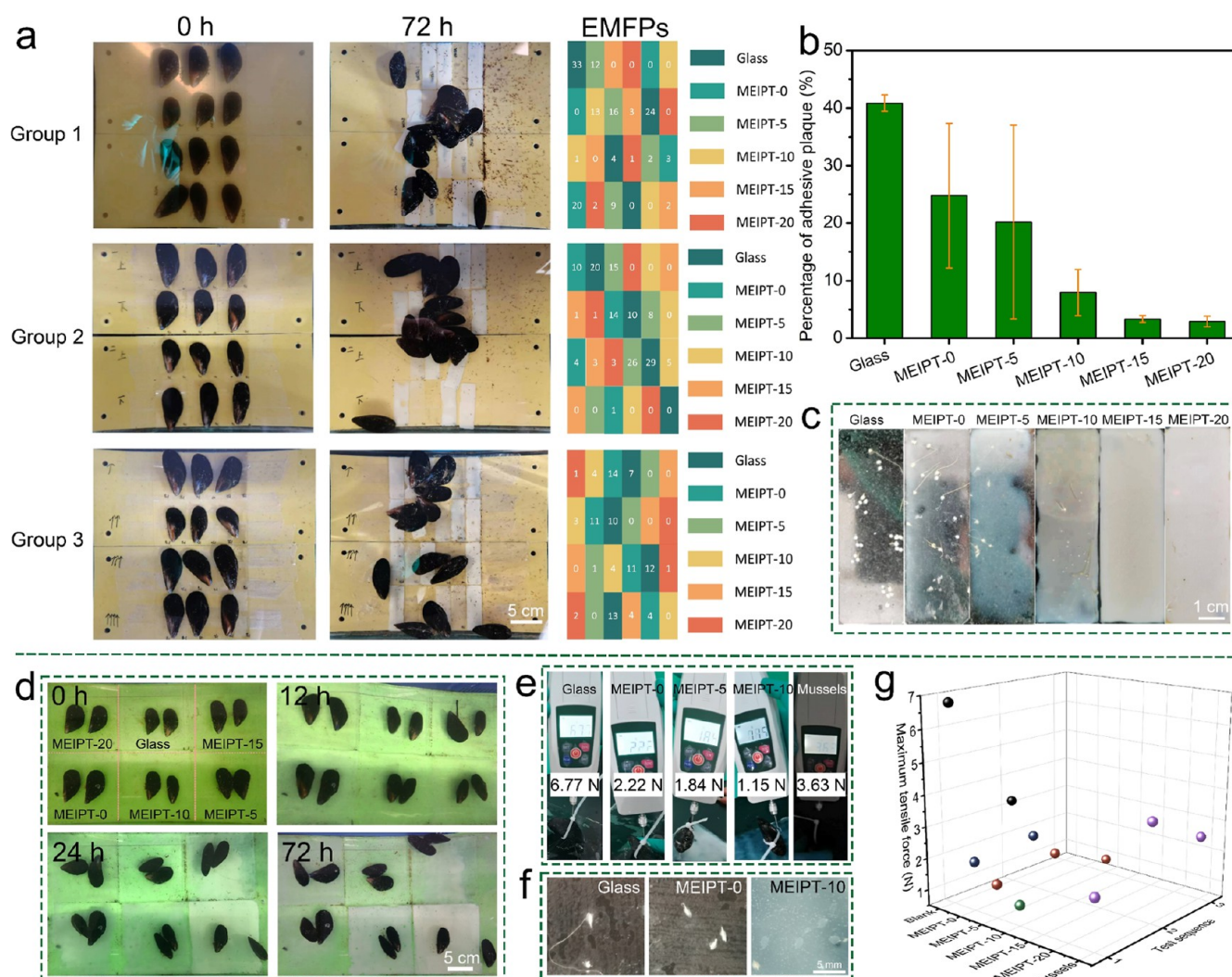


Figure 7. Mussels multichoice assay and settlement results of MEIPT- x coatings. (a) The random distribution of mussels on the coating surface and the position images after 72 h in three parallel experiments, as well as the corresponding number of plaques adhered to different coating surfaces; (b) proportion of mussel plaques adhered to different coating surfaces; (c) photographs of plaques on different samples; and (d) evaluation of adhesive strength of mussels. The results of placing mussels on a 15 cm $\times$ 15 cm coating and their selective adhesion during 72 h period; (e) the digital tension meter measures the maximum tensile force required to remove mussels; (f) optical photograph of mussel residue peeled off from glass and coating; and (g) the maximum tensile force of mussels on each coating obtained from three mussel removal experiments. Note: the error bars represent standard deviations.

absorbance of the *N. closterium* diatom progressively diminishes over time, suggesting a gradual decline in the diatom activity. This result can be attributed to TL acting on certain enzymes or proteins within diatom cells, which inhibits the key photosynthetic pathway. This inhibition affects PSII electron transfer, resulting in the decomposition of chlorophyll in the cells and ultimately causing growth retardation or even death in diatoms.^{35–37} To more effectively illustrate the inhibitory effect of TL on diatoms, the results of toxicity tests conducted over a two week period, examining various concentrations of TL on *N. closterium*, are presented in Figure 6h,i. The concentration of TL ranges from 20 to 50 mg/L, which exerts an inhibitory effect on the proliferation of diatoms. In addition, a concentration exceeding 50 mg/L has a lethal effect on diatoms. The incorporation of TL as an environmentally friendly algicide into SPCs presents significant potential for marine antifouling applications.

Mussels are often considered important pollutants and are more difficult to remove.^{38,39} Figure 7a shows the adhesion

and migration processes of mussels on randomly distributed coating surfaces. It was observed that after 72 h, the mussels selectively left the surfaces of MEIPT-15 and MEIPT-20, subsequently attaching and stabilizing themselves in suitable positions. The numbers on the mosaic experimental board indicate the number of mussel byssus plaques present on the coating surface after 72 h. Figure 7b presents the statistical results from three groups of mussel selectivity experiments. The proportions of plaques observed on the surfaces of glass and MEIPT-0, MEIPT-5, MEIPT-10, MEIPT-15, and MEIPT-20 are 40.9%, 24.8%, 20.2%, 7.9%, 3.3%, and 2.9%, respectively. Compared to coatings with high TL content, mussels prefer to adhere to the surfaces of samples that lack TL or contain low levels of TL (Figure 7c). To further investigate the effect of scaling up the coating on mussel adhesion, first apply the coating onto a 15 cm $\times$ 15 cm glass surface. Then, 12 mussels were placed in a regular pattern on a randomly arranged composite sample (Figure 7d). Over the next 72 h, the mussels exhibited movement and selective adhesion,

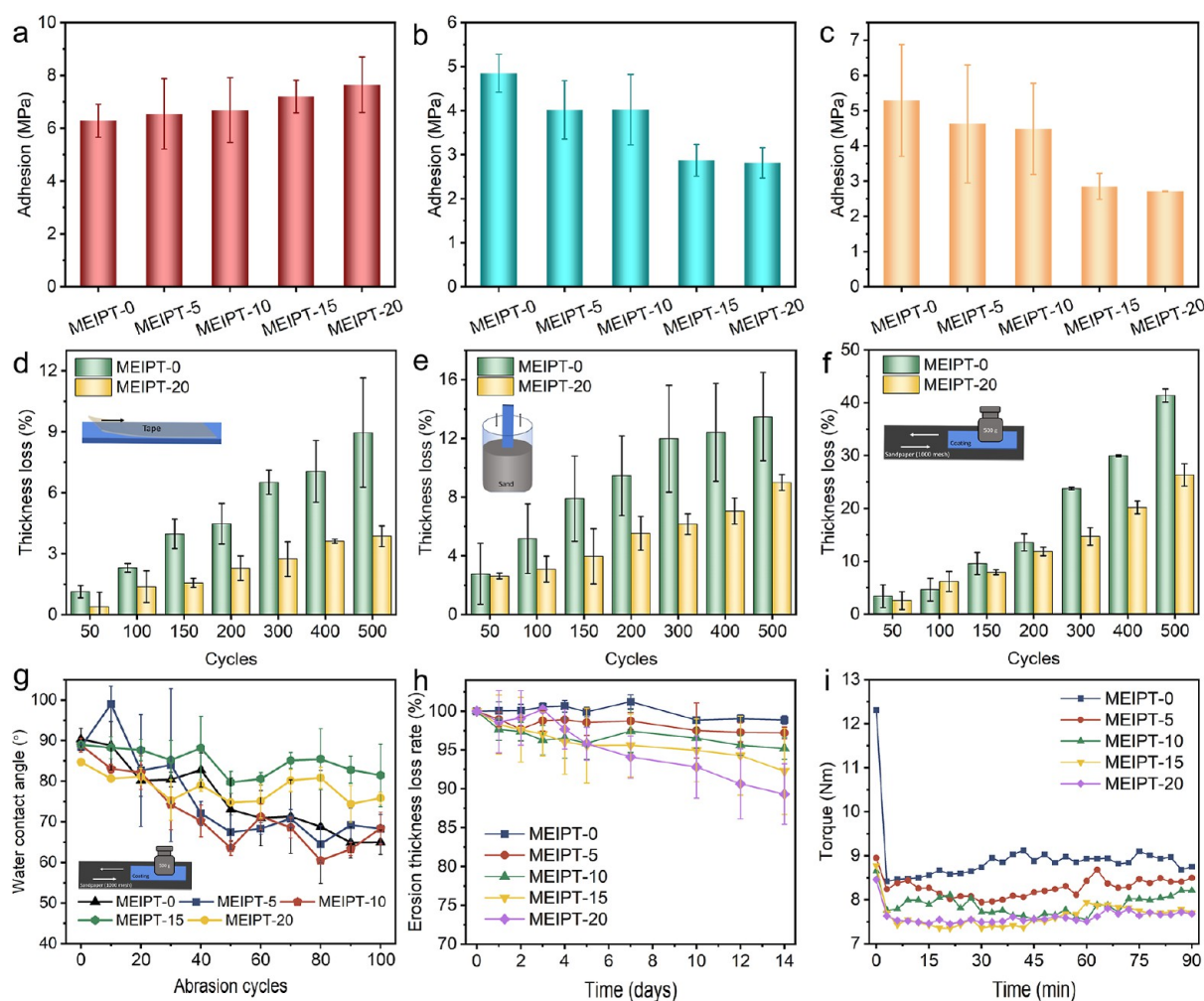


Figure 8. The adhesion of MEIPT-*x* coatings to (a) steel, (b) epoxy, and (c) glass substrate; thickness loss rate after (d) tape peeling, (e) sand abrasion cycles, and (f) sandpaper abrasion cycles on MEIPT-0 and MEIPT-20 surfaces. (g) Contact angle changes of MEIPT-*x* coatings after different wear cycles; (h) thickness reduction rate of MEIPT-*x* coating erosion at 600 rpm water flow. (i) Evaluation of drag reduction performance of MEIPT-*x* coatings under 600 rpm water flow. Note: the error bars represent standard deviations.

further confirming their propensity to escape from samples containing TL and adhere to glass and other mussel surfaces. In addition, the samples were taken out, and the mussels were secured with straps. The strength of the mussel foot line or adhesive spots was tested using a digital tension meter. The adhesive strength of the mussels attached to the sample surface was determined by recording the maximum tensile force during the pull-out process (Figure 7e). The maximum tensile force exerted by mussels on glass is 6.77 N, while the tensile force between mussels is 3.63 N. Additionally, the maximum tensile force on MEIPT-10 is 1.15 N. Notably, no mussel adhesion was observed in the three parallel experiments conducted with MEIPT-15 and MEIPT-20 (Figure 7g). This finding indicates that the coating surface significantly reduces mussel adhesion through surface self-renewing and release of TL. Figure 7f also illustrates that foot filaments and plaques remain on the glass surface, whereas the MEIPT-10 surface displays the imprint only after removal of the mussel plaques. This further demonstrates that the synergistic antifouling strategy of attack and defense employed by the MEIPT-*x* coating can inhibit mussel adhesion.

3.5. Mechanical and Drag Reduction Properties. The mechanical strength of the coating and its adhesion to the

substrate are essential prerequisites for the overall performance of antifouling coatings. Figure 8a–c shows the adhesion strength of copolymers to different substrates. The adhesion strength of MEIPT-*x* to the steel substrate increases with the increase of TL content, and the adhesion of MEIPT-20 to steel substrate even reaches 7.6 MPa. This is attributed to the coordination bond formed between the heteroatoms in the benzothiazolinone of the coating and the iron ions on the steel surface, which enhances adhesion to the steel. In addition, as the TL content increases, the adhesion of the MEIPT-*x* coating to epoxy and glass plates gradually decreases, which may be related to the spatial hindrance of TL. However, the adhesion of the MEIPT-20 coating to both substrates still remained at 2.8 and 2.7 MPa, respectively. The adhesion of MEIPT-*x* coating to all substrates is greater than 1.5 MPa, which meets the requirements for use in complex natural marine environments.⁴⁰ The mechanical surface stability of the MEIPT-20 coating was systematically evaluated by using tape peeling, sand wear cycles, and sandpaper abrasion tests. As shown in Figure 8d, one cycle is defined as a tape adhesion-peeling process. Thickness loss measurements on the coating surface are taken every 10 cycles, with 5 measurement points per assessment. After the tape peeling test, the thickness loss of

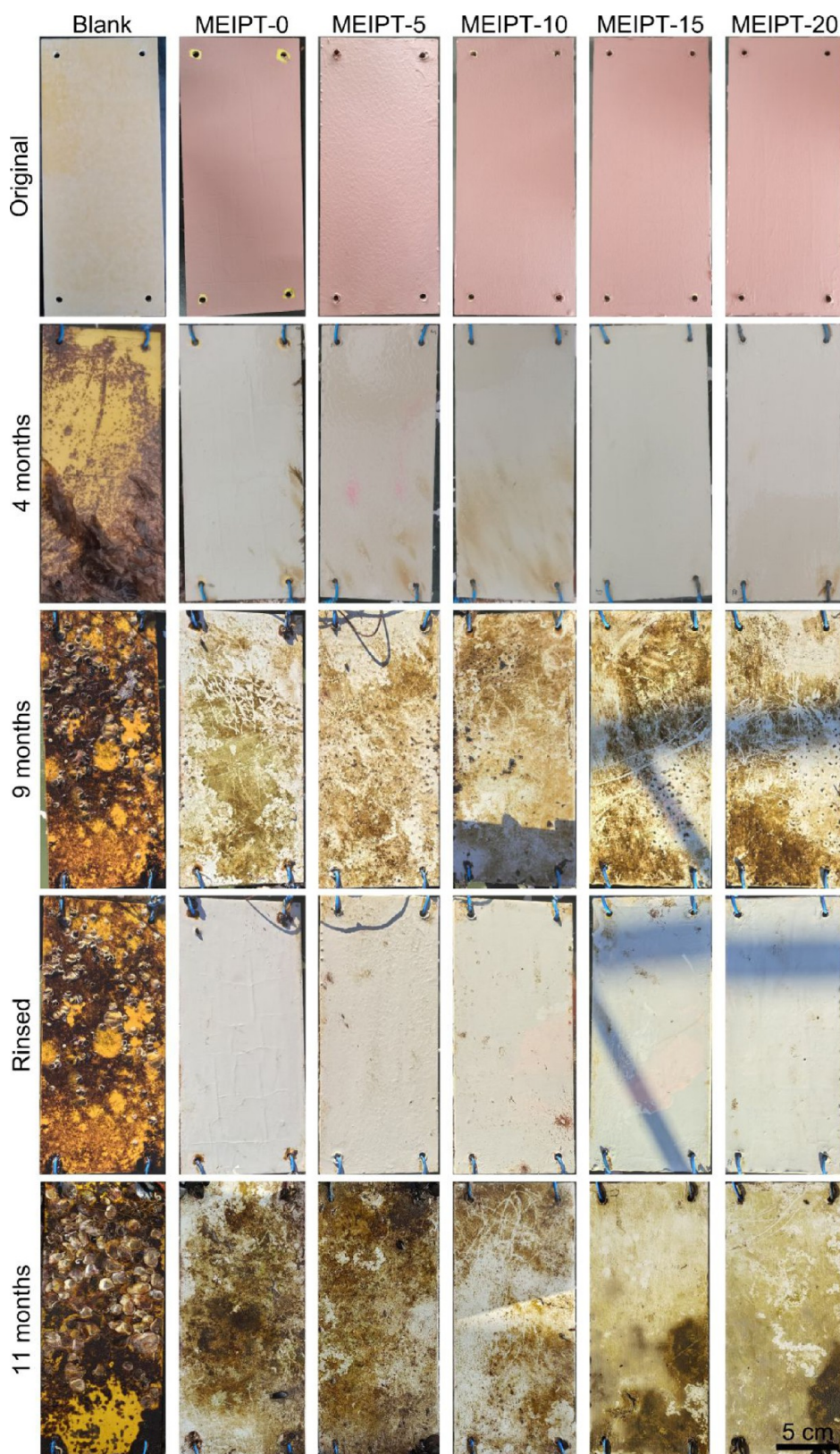


Figure 9. Photographs of a blank epoxy plate and MEIPT- x coatings submerged in the sea area of Dalian of China for 11 months (the surface is subjected to high-pressure rinsing after 9 months).

the MEIPT-20 coating was significantly lower than that of the MEIPT-0 coating. The sand wear cycle test was conducted by inserting the coating into a sand pile and lifting it to complete one sanding cycle (Figure 8e). After 500 cycles, the thickness

loss of the MEIPT-0 coating was 13.5%, whereas the MEIPT-20 coating exhibited a thickness loss of only 8.9%. We employed more rigorous methods to further investigate the mechanical stability of the coatings. Each coating was placed

on 1000 mesh sandpaper and subjected to a 500 g load. The coating was then moved 10 cm across the sandpaper surface to complete one wear cycle (Figure 8f). Experimental results showed that after 500 wear cycles the thickness loss of the MEIPT-0 coating was as high as 41.4%, whereas the MEIPT-20 coating decreased by only 26.3%. The relationship between WCA and wear length is shown in Figure 8g. As the number of wear cycles increases, the WCA of the MEIPT-*x* coating gradually decreases. After 100 wear cycles, the contact angles of MEIPT-0, MEIPT-5, MEIPT-10, MEIPT-15, and MEIPT-20 decrease by 25.4°, 20.1°, 20.5°, 7.5°, and 8.9°, respectively. The initial contact angle of the MEIPT-*x* coating system is below 90°, indicating a low density of hydrophobic functional groups on the coating surface. After the wear cycle test, the hydrophobic groups on the surface of the MEIPT-0 coating were severely damaged. Furthermore, according to Wenzel's equation, the increased roughness of hydrophilic surfaces after wear enhances their hydrophilicity. For MEIPT-20, its high content of benzene and thiazolinone rings increases the rigidity of the coating, making the surface more wear-resistant and resulting in a smaller decrease in the contact angle. Figure 8h illustrates the trend of the coating thickness variation under water flow erosion at 600 rpm. The thickness loss rate of MEIPT-20 is the highest, at 10.7%. Although MEIPT-20 exhibits good mechanical stability, the hydrolysis rate of the coating's side chains accelerates under dynamic water flow erosion. This leads to an increased self-renewal rate of the coating surface and a greater loss of coating thickness. Navigation resistance directly impacts the speed, energy consumption, and carrying capacity of ships. The application of specialized drag-reducing materials or coatings can diminish surface friction, thereby further decreasing navigation resistance.⁴¹ Figure 8i shows the relationship between times and torque at 600 rpm for different coatings. A large torque indicates a high water resistance. The drag reduction rate is calculated according to eq 1

$$\text{DR\%} = \frac{T_0 - T_c}{T_0} \times 100 \quad (1)$$

where T_0 is the torque measured on the MEIPT-0 surface and T_c is the torque measured on the other coating surfaces. As the TL content increases, the water resistance gradually decreases. The maximum drag reduction rate of MEIPT-20 is 12.2%, indicating that the coating has a drag reduction effect. The enhanced drag reduction performance of MEIPT-20 is attributed to the formation of a stable and continuous hydration layer on its surface following the hydrolysis of numerous side chains, which reduces friction between water flow and the coating.

3.6. Antifouling Evaluation after Surface Self-Renewing. Figure S3 illustrates the evaluation of antibacterial performance for the MEIPT-*x* coating surface following water flow erosion and self-renewing. The antibacterial efficiency of surface renewed MEIPT-20 against *E. coli* and *S. aureus* increased from 57.8% and 62.0%, to 81.4% and 90.8%, respectively. The activity of *P. xiamenensis* marine bacteria is relatively low. Therefore, *Pseudoalteromonas marines*, which exhibit marine bacteria of higher activity, are employed to assess the antibacterial performance after surface self-renewing. Nevertheless, its antibacterial efficiency against marine bacteria still reached 95.7%. The enhancement of antibacterial performance can be attributed to two factors: first, the water

flow eroded hydrophilic surface is more likely to form a hydration layer, which hinders contact between bacteria and the coating. Second, the water flow eroded surface increases the exposure area of the antibacterial agent TL, significantly improving both the antibacterial rate and bacterial mortality.

Figure S4 shows the evaluation of the anti-diatom performance of the MEIPT-*x* coating surface after water erosion and self-renewing. It can be observed that as the TL content increases, the number of *Ha.* and *N. closterium* diatoms on the coating surface gradually decreases. After self-renewal of the MEIPT-20 coating surface, the density and diatom resistance ratio of *Ha.* after 7 days of cultivation were 27 cell mm⁻² and 98.1%, respectively. This is similar to the inhibition effect of the *Ha.* diatom before surface renewal. This is attributed to the repellent and killing effects of the antifouling active sites on the surface of the coating on benthic diatoms. However, planktonic diatoms move solely with the flow of water. Therefore, the increase in the coating surface roughness following water flow scouring will enhance the contact area between planktonic diatoms and the coating, resulting in a higher adhesion density of *N. closterium* diatoms.

3.7. Real-Sea Antifouling Assessment. To further investigate the antifouling performance of the MEIPT-*x* coatings in real marine environments, we conducted antifouling experiments in the sea area of Dalian, China (Figure 9). It can be observed that after 4 months (from January to May), a significant amount of sludge and seaweed accumulated on the epoxy plate. In contrast, the MEIPT-*x* coatings surface exhibits only negligible sludge. This difference is attributed to the release of antifouling agents from the coating as well as the slow growth of marine organisms during colder seasons. From May to October, as the temperature increases, marine organisms enter their peak growth season. During this time, a significant number of large diatoms, barnacles, shellfish, and other marine organisms attach themselves to the epoxy plate. The MEIPT-0 coating has a large area of biofilm and one shellfish attached to its surface. In contrast, the other MEIPT-*x* coatings only have some sludge adhering to their surfaces. In addition, despite high-pressure water flushing, the fouling organisms on the epoxy plate continue to adhere firmly to the surface, but the surfaces of MEIPT-*x* coatings are once again refreshed. This indicates that despite the peak growth season for fouling organisms, the MEIPT-*x* coatings maintain excellent antifouling performance and can self-renew on the surface under the erosion of water flow, further preventing fouling organisms from adhering. After an additional 2 months of immersion, the adhesion area of large shellfish to the epoxy plate reached 80%. The MEIPT-0 and MEIPT-5 coatings adhered to a significant amount of sludge and a small number of fouling organisms, whereas the coating surfaces of MEIPT-10, MEIPT-15, and MEIPT-20 exhibited only minimal sludge. In summary, the MEIPT-*x* coatings demonstrate a long-lasting antifouling performance and stability, indicating their practical application value.

4. CONCLUSION

In summary, inspired by the dual antifouling strategy of "passive defense and active attack" of coral, the eco-friendly self-renewing antifouling coating was developed. In static simulation experiments, the MEIPT-*x* coating demonstrated an excellent self-renewing performance. Throughout the testing process, the weight loss rate and antifouling agent release rate of the coating rapidly stabilized, which is advantageous for

ensuring that the antifouling effect of the agent remains uniform and long-lasting. In the static antifouling evaluation conducted in the laboratory, the MEIPT-*x* coatings demonstrated effective antibacterial properties, diatom eradication capabilities, and mussel-repellent effects. After 14 days of scouring, the antibacterial and diatom resistance of the coating continued to improve, indicating that the antifouling performance of the coating is both long-lasting and stable. It is noteworthy that during the 11 month evaluation of real-sea antifouling performance, there was no attachment or colonization of fouling organisms on the surface of the MEIPT-20 coating, indicating its long-lasting and effective antifouling effect. In addition, the coating exhibits good drag reduction performance, wear resistance, and strong adhesion to the substrate. The environmentally friendly MEIPT-*x* coating developed in this study is consistent with the current trend of creating long-lasting, efficient, and environmentally sustainable antifouling coatings.

■ ASSOCIATED CONTENT

Data Availability Statement

Data will be made available on request.

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsami.5c24167>.

Experimental materials; application method and schematic diagram of MEIPT-*x* coatings; experimental characterization; synthesis formulas of MEIPT-*x* coatings; nuclear magnetic resonance hydrogen spectrum of TLAMA; antibacterial images of the MEIPT-*x* coatings after dynamic scouring of 600 rpm; antidiatom images of the MEIPT-*x* coatings after dynamic scouring of 600 rpm; and environmentally friendly performance of MEIPT-*x* coatings (PDF)

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Author Contributions

Z.Z.: conceptualization, methodology, data curation, and writing—original draft. **J.Z.:** investigation. **R.C.:** writing—reviewing and editing. **G.S.:** project administration. **Y.Z.:** software. **J.Y.:** investigation. **J.L.:** investigation. **R.L.:** writing—reviewing and editing, formal analysis, and funding acquisition. **J.W.:** funding acquisition.

Notes

The authors declare no competing financial interest.

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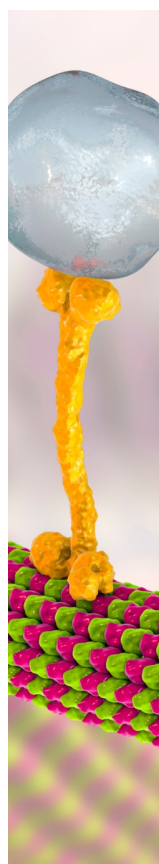
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